FOOD WASTE MONITORING SYSTEM

Project Explanation and Algorithm

1. Assigned Syllabus Matter

Python solutions using only the assigned syllabus: computational thinking, algorithms, flowcharts, pseudocode, variables, data types, operators, conditional statements, loops, strings, pattern generation, lists, nested lists, tuples, tuple operations, and unpacking.

2. Aim

To monitor daily food wastage by calculating the percentage of leftover meals and identifying the days on which wastage exceeds a user-defined acceptable limit.

3. Description

The program accepts the number of days and an acceptable wastage percentage. For each day, it reads the number of meals prepared, meals served, and leftover meals. It treats the number of leftovers as the wastage, calculates the wastage percentage relative to meals prepared, and compares that percentage with the acceptable limit. Days exceeding the limit are stored in a list and displayed in the final report.

4. Inputs

- **Number of days:** Total days to monitor.
- **Acceptable wastage percentage:** The maximum permitted wastage percentage.
- **Meals prepared:** Total meals prepared on each day.
- **Meals served:** Meals served on each day (read as input but not used in the calculation).
- **Leftover meals:** Number of meals remaining after service.

5. Variables and Data Structures

days: Integer storing the monitoring period.
acceptable_limit: Float storing the permitted wastage percentage.
high_waste_days: List containing day numbers that exceed the limit.
day, prepared, served, leftovers, wastage: Integer variables used during daily processing.
wastage_percent: Float storing the calculated percentage.

6. Formula

Wastage = Leftover meals
Wastage Percentage = (Wastage / Meals Prepared) × 100

A day is added to **high_waste_days** only when its wastage percentage is strictly greater than the acceptable limit. A percentage equal to the limit is not considered an exceeded day.

7. Algorithm

1. Start.
2. Read the number of days and acceptable wastage percentage.
3. Create an empty list named high_waste_days.
4. Repeat for each day from 1 through the given number of days.
5. Read meals prepared, meals served, and leftover meals.
6. Set wastage equal to leftover meals.
7. Calculate wastage percentage using the formula.
8. Display the wastage percentage.

9. If the percentage is greater than the acceptable limit, append the day number to high_waste_days.
10. After all days are processed, display the food waste report.
11. If the list is empty, display that no day exceeded the limit; otherwise display each stored day number.
12. Stop.

8. Pseudocode

```
BEGIN
    INPUT days
    INPUT acceptable_limit
    high_waste_days = empty list

    FOR day = 1 TO days
        INPUT prepared
        INPUT served
        INPUT leftovers

        wastage = leftovers
        wastage_percent = (wastage / prepared) * 100
        DISPLAY wastage_percent

        IF wastage_percent > acceptable_limit THEN
            APPEND day TO high_waste_days
        END IF
    END FOR

    IF high_waste_days is empty THEN
        DISPLAY "No days exceeded the acceptable limit"
    ELSE
        FOR each day IN high_waste_days
            DISPLAY day
        END FOR
    END IF
END
```

9. Output

The program displays the wastage percentage for every day, followed by a report. If no day exceeds the acceptable percentage, it displays a message stating so. Otherwise, it lists the day numbers that exceeded the limit.

10. Important Notes

• The calculation uses leftover meals as wastage; meals served are collected but are not used in the formula.
• The comparison is strict (>), so equality with the acceptable limit does not trigger a warning.
• Meals prepared must be greater than zero to avoid division by zero.
• The input values should be valid non-negative counts, and the number of days should be a positive integer for a meaningful report.

11. Conclusion

The Food Waste Monitoring System provides a simple way to track leftover meals over multiple days. By calculating daily wastage percentages and identifying days above the acceptable threshold, it helps highlight when food preparation may need to be adjusted.